Samruddhi Oke

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EDUCATION

Cornell University, GPA: 3.6 / 4.0 **Expected Graduation: May 2028**

Bachelor of Science: Computer Science (Honors); Specialization: ML, AI | ***Honors:*** TATA Scholar (Top 1%)

Teaching Assistant: Large Language Models (SP26), Machine Learning (FA26), Embedded Systems (FA26)

Relevant Courses: Analysis of Algorithms, Systems for Large-Scale ML, Data Structures, Object-Oriented Programming, Data Science, Artificial Intelligence, Quant Finance, Discrete Mathematics

RESEARCH EXPERIENCE

Cornell University Ann S. Bowers College of Computing and Information Science January 2025 – Present

Student Researcher under Prof. François Guimbretière | PilotProxy Under Review: **CHI' 27**

- Built multimodal sensor-fusion models (PyTorch, OpenCV, CUDA) holding ±0.5 m localization through 30% sensor dropout.
- Optimized a CUDA CV pipeline to 30 FPS with <50 ms YOLOv8 inference for real-time autonomous robotic navigation.
- Engineered C++ and ROS navigation software for dynamic A* path planning, reducing manual operator intervention by 67%.
- Designed offline and online evaluation harnesses to isolate sensor, model, and navigation failures before real-world deployment.

PROFESSIONAL EXPERIENCE

I Can't Tell Anyone May 2026 – Present

Founder & Applied AI/ML Engineer

- Developed, deployed and maintain a full-stack AI social media site (React/Next.js, Python, FastAPI) for 160+ users end to end.
- Built an asyncio WebSockets with Redis Pub/Sub and PostgreSQL routing chat, feeds, and moderation across 5K+ live streams.
- Achieved <25 ms inference by fine-tuning a PyTorch/Hugging Face classifier with LoRA and deploying ONNX INT8 inference.
- Deployed Llama 3.2 inference services with Triton and Docker, isolating sensitive user data within controlled infrastructure.

Cornell Algorithmic Trading Club January 2026 – Present

Director of Data & Infrastructure

- Led AWS/GCP cloud infrastructure supporting 20+ researchers and a $150K fund, serving REST APIs and shared data services.
- Trained CNN and NLP classifiers (TensorFlow, scikit-learn, NumPy) over 15K+ SEC filings, cutting ingestion time 40%.
- Designed real-time SQL/NoSQL storage across PostgreSQL, Redis, and Firebase for durable, cached, and live-state market-data.
- Standardized PR review, testing, data versioning, AWS access controls, and technical runbooks across shared infrastructure.

ZetaChain May 2025 – July 2025

Software Engineering Intern

- Cut distributed Node.js API latency 56% via cross-chain RPC batching, Redis caching, and PostgreSQL query optimization.
- Reduced schema errors 25%, saving 20+ debugging hours/week with reusable Zod/TypeScript validation across 20K+ LOC.
- Built an MCP-enabled TypeScript CLI with Claude Code for runtime diagnostics and RPC tooling, adopted by 8 engineers.
- Expanded automated testing to 90% coverage and Dockerized Go services on Linux, reducing CI/CD failures by 30%.

Cornell Blockchain Club September 2024 – Present

Full-Stack Software Engineer & Recruitment Lead

- Led 5 engineers through sprint planning, API design, PR reviews to ship a React/Next.js web and mobile gig marketplace.
- Developed async FastAPI services (SQLAlchemy pooling, PostgreSQL) sustaining 10K+ concurrent requests in load tests.

PROJECT EXPERIENCE

Legal Eagle – *Multimodal Information Retrieval & Generative AI System* | *Python, LangGraph, LangChain, Pinecone, JavaScript, Redis*

- Designed a multimodal RAG retrieval architecture turning legal filings into dual-host podcasts, cutting LLM token costs 52.5%.
- Reduced generation hallucinations 38% using asynchronous multi-agent pipelines, grounding checks, and automated evaluation.

Clout Trader – *Influencer Prediction Market* *(2nd Place at UPenn Blockchain Hackathon)* | *Solidity, Python, Next.js, NLP, TailwindCSS*

- Processed 12K+ tweets/day to NLP signals for a Solidity/Next.js prediction market executing $48K+ simulated on-chain trades.

Bitcoin Price Prediction – *Production ML & Streaming Pipeline* | *Python, XGBoost, Kafka, FastAPI, Prometheus, REST APIs*

- Architected a real-time Kafka/PostgreSQL pipeline with <5s streaming latency processing large-scale market data.
- Trained an XGBoost/LightGBM ensemble on 40+ features with <150ms FastAPI inference & Prometheus model drift alerts.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, C, Go, TypeScript, JavaScript, C#, SQL, Solidity, Rust, R, MATLAB, Kotlin
- **Full-Stack:** React.js, Next.js, Node.js, FastAPI, Flask, SQLAlchemy, Tailwind CSS, REST APIs, CI/CD, HTML/CSS
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face, LangChain, LangGraph, OpenCV, CUDA, Triton, JAX
- **Infra & Tools:** ROS, AWS, GCP, Docker, Git, Redis, PostgreSQL, Linux / Unix, Firebase, Claude Code, Cursor